

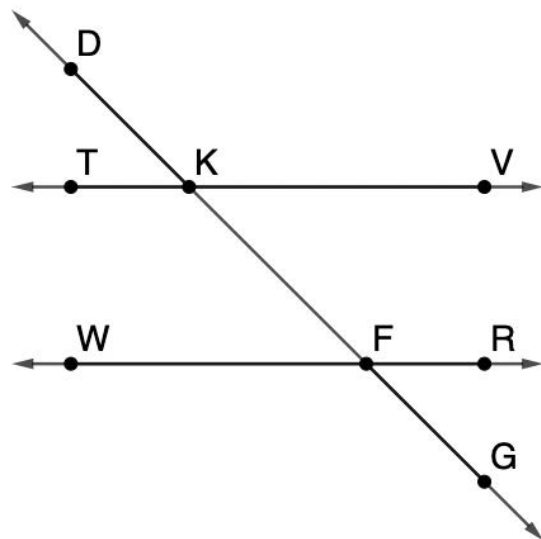
Geometry
Unit 3
Lesson 9 Homework

Name _____
Date _____
Period _____

Complete the two-column proof.

Given: $\angle TKF \cong \angle KFR$

Prove: $\angle VKD \cong \angle WFG$



Statement	Reason
$\angle TKF \cong \angle KFR$	Given
1.	2.
3.	4.

Write the statements and reasons in the correct order in the two-column proof.

$\angle TKD \cong \angle GFR$

Supplements of congruent angles are congruent.

$\angle VKD \cong \angle WFG$

If two angles form a linear pair, then they are supplementary

$\overleftrightarrow{TV} \parallel \overleftrightarrow{WR}$

If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.

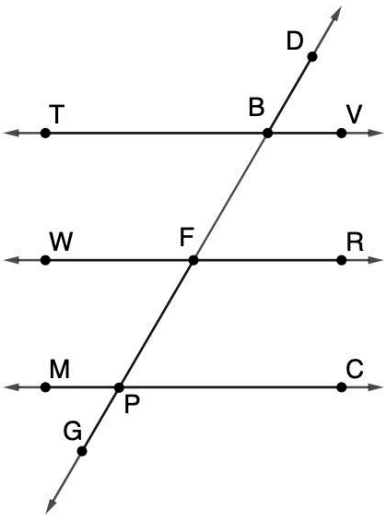
$\angle GFR$ and $\angle WFG$ are supplementary

If alternate interior angles are congruent, then the two lines that are intersected by a transversal are parallel.

Complete the two-column proof.

Given: $\angle DBV \cong \angle MPG$ and $\angle TBD \cong \angle WFB$

Prove: $\angle RFP \cong \angle MPF$



Statement	Reason
$\angle DBV \cong \angle MPG$	Given
5.	5.
6.	6.
7.	7.
8.	8.
9.	9.
10.	10.

Write the statements and reasons in the correct order in the two-column proof.

$\angle PFR \cong \angle MPF$	Given
$\angle TBD \cong \angle WFB$	Transitive Property of Congruence
$\angle PFR \cong \angle TBD$	If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.
$\angle TBD \cong \angle MPF$	If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
$\overleftrightarrow{TV} \parallel \overleftrightarrow{MC}$	If alternate exterior angles are congruent, then the two lines that are intersected by a transversal are parallel.
$\overleftrightarrow{TV} \parallel \overleftrightarrow{WR}$	If corresponding angles are congruent, then the two lines that are intersected by a transversal are parallel.